

Reflection groups for learning professional skills for computer science, with transfer beyond the course

Firstname Lastname 1
email1@institution2.edu
Institution 1
City, State, Country

Firstname Lastname 2
email@institution2.edu
Institution 2
City, State, Country

Firstname Lastname 3
email6@institution2.edu
Institution 1
City, State, Country

Firstname Lastname 4
email6@institution2.edu
Institution 1
City, State, Country

Firstname Lastname 5
email6@institution2.edu
Institution 1
City, State, Country

Firstname Lastname 6
email6@institution2.edu
Institution 1
City, State, Country

ABSTRACT

In education and society, it is important to reflect in order to improve ourselves and use our tools more critically and ethically. However, there is a persistent gap in computing education on teaching these and other professional skills for lifelong learning and communication, with scarce course and faculty time available. The key insight of our work is that it may be easier to teach students to support each other in reflecting and improving, when we make that the primary goal of the group (instead of the group goal being a software project or some other task). We created these **reflection groups**, loosely adapting agile retrospective practices; groups primarily met weekly outside of class time. We incorporated them into our computer science ethics course (an in-the-major course that satisfies ABET accreditation requirements). Based on a qualitative thematic analysis of end-of-semester group reflections, students learned to open up and be vulnerable to a dramatic degree. They improved at professional skills such as communication and listening. They adopted iterative improvement as a habit and became more honest with themselves. Some even transferred those skills and reported positive impacts their relationships outside the course, even in their families. Better understanding how and why this happened, and how to replicate it, is important future work. Reflection groups require minimal instructor time, offering a practical way to develop professional skills. Reflection groups also build connections and community among students, which is even more important as AI is disrupting traditional peer learning communities.

CCS CONCEPTS

• **Social and professional topics** → **Computing education**.

KEYWORDS

mixed methods, qualitative, generative AI, qualitative

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1 INTRODUCTION

Reflection, iterative improvement, and supporting others are essential professional skills for computing, for lifelong learning and ethical practice. These professional skills and dispositions are a part of ACM CS2023 [36] and found in research on industry [19, 20, 28]. The skill and idea of improving our ability to improve is historically a foundational part of computing culture, originating with Douglas Engelbart who described it as the key enabler of the success of NLS and the “mother of all demos” in 1968, which included the invention of collaboration features, links, the mouse, and the layered domain specific language architecture [23, 24].

In prior work in computing education, professional skills are often designed to be learned via team projects building software or producing some output [28, 43]. However, this is insufficient as there is a persistent computing academia-industry gap on learning professional skills; review and aggregate studies found continuous and lifelong learning [28, 29], and oral and written communication [45] as top skill deficits. A large 2025 ITiCSE working group found “...lifelong learning and professionalism are under-emphasised.” [28]. Valstar et al. found computing faculty think more resources are needed to teach and give feedback on professional skills [53], but faculty are chronically under-resourced.

Our work addresses this professional skills and resourcing gap with a relatively low instructor workload assignment for learning professional skills, including lifelong learning and oral communication, based on carefully scaffolding peer observation and peer learning, via regular outside-of-class weekly reflection meetings with agendas and goal-setting. We contribute a novel instructional design for an ongoing assignment and group learning environment for learning reflection, iterative improvement, and supporting others, via **reflection groups**. These groups did not overlap with a team project, making the group’s undivided purpose the growth and well-being of the team members. Our design was inspired by the practice of sprint retrospectives from agile software development [51], but adapts that pattern outside a group project setting.

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We provided scaffolding via a weekly meeting agenda. After each meeting, students turned in individual weekly reflections, graded for completion; attendance was not directly graded.

We implemented and studied these reflection groups in a three credit in-the-major computer science ethics course, taught synchronously online only. We also frequently used these groups during class for active learning and discussion. We asked the following research questions:

- (1) **RQ1** How might we design a collaborative learning assignment and environment for learning reflection, iterative improvement, and supporting others in iterative improvement and continuous learning?
- (2) **RQ2** What categories of learning and effects were reported from the reflection groups by students?
- (3) **RQ3** According to student reflections, what helped students a) improve, and b) improve their ability to improve as a reflection group?

To answer RQ1, we present our reflection group design in Section 3. To answer RQ2 and RQ3, we conducted a qualitative thematic analysis on students' end of semester retrospective reflections.

2 RELATED WORK

Many learning interventions include a group component, with the primary goal of learning the course's material, but in computing education we are unaware of published work on regular non-project team reflection groups in general (and in particular with our design for reflection groups). Reflection in computing often focuses on technical learning [43], where our work focuses on professional skills (students did in practice also choose to reflect on technical topics). Beyond computing education, in a literature review on group reflection research across disciplines, they found a gap in qualitative research on how members perceive the group's affordances [55], which our RQ3 addresses.

The closest work to our non-project team reflection groups is in teacher education and professional medical education. Group reflection is relatively rare in practice and in research studies outside of those fields [13, 55]. Several teacher reflection group examples related to computing can be found in [7], such as Ryoo et al. where teachers reflect on communication and other challenges in transitioning to an inquiry-based classroom culture [46]. Within medical education, common group designs include debriefing after a shift and reflecting on challenging cases [41]. In comparison, our groups have more student agency, via a broader reflection scope including inside and outside of the course (inside the domain of practice and outside), and a light post-meeting individual written reflection.

Some work in computing has facilitated more inclusive spaces for learning and feedback, such as peer-led team learning (PLTL) or supporting students forming study groups for homework, but not for learning general reflection and iterative improvement skills. PLTL specifically refers to trained peer facilitators who are not taking the course acting similar to TAs [21, 27, 50, 54]. Other work addresses scheduling and other barriers to participation in study groups, but not designed or evaluated for learning reflection and iterative improvement skills [12, 33].

There is a small body of work within computing education on student project teams reflecting to improve. Some work designs

more scaffolding to improve sprint retrospectives, and found improvements such as Tariq et al. that augmented a stop start continue reflection form [52]. Some work analyzes project team sprint retrospectives [32] or reflection on design projects [22]. Outside of computing, there is more work, such as Schurmann et al.'s longitudinal qualitative study found project team reflection scaffolded with behavioral rating scale helped learning to collaborate on a project [47]. Other work has found virtual teams can be non-inferior to in-person teams [35].

Some computing education work has tried to teach reflection or implement processes that include reflection, but not via regular non-project team reflection groups where peer observation and feedback drive learning. For example, some courses use a studio approach with a high staff to student ratio where teachers explicitly model and guide reflective critique sessions [10, 14, 44]. Babb et al. present guidelines for integrating reflection [5]. Courses integrate reflection [18, 26], such as Burden et al. designed software engineering (SWE) process reflections into a SWE course [11]. A few longitudinal qualitative works include computing student reflections [42], and Barr, Andrei, and others found that reflection skills took years to develop even with workplace-based learning [3, 6].

Courses teaching professional skills in computing have focused primarily on communication, such as explicitly modeling interpersonal skills, e.g., communicating perceptions with empathy [8].

In computing ethics education, approaches include consistent breakout discussion groups [39]. Another somewhat similar work is a tutorial-like course involving ethics with 6 to 7 students plus a tutor "who also has a more general pastoral responsibility for those students for the year." described in a couple paragraphs within [2]. Other courses have incorporated individual reflection [9, 38], such as logs or journals [48], self-evaluation with respect to ethical guidelines [17], and peer feedback on assignments [43]. Group class discussion of ethics is common and is often in the context of ethical frameworks and applying them to cases; see Brown et al. for a comprehensive review on this and other teaching for ethics [9].

3 DESIGN OF REFLECTION GROUPS AND COURSE CONTEXT

Here is a short summary of the reflection group assignment; each part is described in more detail later in this section. Reflection groups were created in the third week of class using survey data that asked students about scheduling constraints and identity through shared gender and disability status. There were nine reflection groups for 38 students. As part of the assignment, each group had a 30-minute weekly reflection meeting online outside of class time. Each meeting had three parts. First, students would share their progress and challenges. Next, they would affirm challenges encountered, encourage each other, and celebrate accomplishments by commenting on progress and perseverance. Last, students would generate ideas for iterative improvement. During meetings students could offer help to group members.

To form critical initial bonds and safety, we started with an in-class group icebreaker, based on an activity from experimental psychology for creating interpersonal closeness among two people [4]. We adapted it by removing questions that might bring up trauma, like "If you could change anything about the way you were raised, what would it be?" and, for the group setting, shortened

questions with time guidelines, for example changing *four minutes* to “Take a *minute* and tell your teammates your life story in as much detail as possible.” Before the activity, we announced in that students could pass if uncomfortable; this is helpful for non-neurotypical and other students. The instructor and TA cycled through the groups and responded to whatever question they were on.

For the 30-minute weekly reflection meetings outside of class, students were provided a meeting agenda with guided questions to support conversation and scaffold learning for foundational professional skills and personal development. Just as in purely technical computer science, sometimes the best solution to a problem is to instead solve a more general problem. Thus, we targeted first general iteration and improvement skills, which then students apply in order to learn other professional skills. Our question design prioritized simplicity and the skill of iterative improvement, while scaffolding a basic reflective learning cycle [34]: 1) What went well for your personal growth and learning this week? 2) What supported your personal growth and learning? 3) What might be improved for your personal growth and learning? The improvement can be anything, inside or outside of classes. 4) Brainstorm possible improvement goals for the next week for your personal growth and learning. As long as a goal is meaningful for you, a goal can be anything, inside or outside of classes, and 5) Discuss choosing an improvement goal for the next week, and why you might choose that goal. As long as the goal is meaningful/important for you and related to some goals you have, the goal can be anything, inside or outside of classes. The broad question scope allows students to choose the computing professional skills most of interest and relevant for them.

After the meeting, students would submit individual private answers to those questions via a form and choose an improvement goal for the next week. These answers were only viewable by course staff, graded for submission only, and the only graded component.

Each reflection group also had a 45- to 60-minute end of semester retrospective meeting outside of class, with a different agenda. Students typed out their answers to the first section on “What did you individually improve over the semester?” by answering these four questions: top improvements, “How did others in the group help you?”, “How did this happen? what caused it? why was it done?”, and “How did this improvement affect you and others?”. They then shared their answers with their group and discussed their answers. Then the group did the same for the next section: improving your individual ability to improve (i.e., getting better at getting better) [24]. Finally, they do that again for the last section: improving their ability to improve as a group [24]. [All of the materials for the reflection groups are available here \(anonymized\).](#)

3.1 Course context

The computer science ethics course covered traditional ethics topics, with direct application to professional situations and students’ lives. For example, a week on machine learning and optimization included discussion of situations where “*you* over-optimized something or have a habit of over-optimizing that has some undesirable consequences.” and “How might you focus on more and/or balance better other important values in a situation like this?”, followed by parallel questions for “a situation where a *technology* over-optimized something...that has some undesirable consequences for you or people you care about.” The reflection groups were intentionally

placed in this course because it recognizes how personal habits impact professional life. The course also serves as the required ethics course in our ABET accredited Computer Science BS degree program, including a writing component. The course followed the normal approach of reading assignments and active in-class learning activities related to the readings.

The reflection groups started one month into the course with about sixty minutes of a class session dedicated to the icebreaker activity. The readings had consistent reflection questions to practice reflecting on learning: “What I learned”, “What I knew, thought, and/or felt before about this”, and “Why this learning was important and/or meaningful for me”. Additional reading-related reflection questions were often added.

4 METHODS

We conducted a qualitative thematic analysis by inductively memoing and categorizing students’ written end of semester retrospective reflections from their final reflection group meeting. We chose this dataset to focus on overall learning and effects as well as experiences at the group and individual level, and for feasibility.

For the analysis, each member of the qualitative analysis team read and memoed the data, generating 583 memos with associated quotes. We then affinity diagrammed those to create higher-level themes for our research questions. We discussed the categories, removed sub-categories with fewer than three quotes, and revised descriptions until we agreed they accurately reflected the supporting quotes. We also returned to the full group retrospective documents to triangulate and check our understanding when unsure. Instead of inter-rater reliability, we used the approach of resolving disagreements through collaborative discussion [30], consistent with acceptable qualitative research practices in computing education (e.g., [25]). Disagreements revolved around coming to agreement that vague quotes did not fall under more specific themes (e.g., “open communication” not under “Communicating struggles”) and merging related themes such as *experimentation* and *trial and error with reflection* into one. Finally, we wrote the results by walking through each theme with sub-themes and example student quotes.

The qualitative analysis team comprised the authors: two students who completed the course, the instructor, and an equal number not involved in the course (two students and an external qualitative computing education researcher).

Our entire study was approved by our institutional ethics review board and followed all required practices for human subjects research. Of 38 students enrolled in Spring 2025, one was inactive, and three opted out of the research (two MS, one BS). Thus, our 34 participants were 22 undergraduate computer science majors and 12 professional MS students in [Anonymized program]. The course was a required computer science ethics course for both programs. The demographics survey was given in the third week of class, see Table 1 below (two research participants did not respond).

Using the demographics survey as stated in the group design section, we matched groups on gender and disability status where scheduling allowed, as demographic similarity in small groups supports psychological safety and willingness to share [16, 49]. This gave one all disabled women group of 4, one mixed gender group of 3, one mixed group of five with 3 men (an evening all professional MS students group). The remaining five all men groups

Gender: Men 28 (88%) / Women^a 3 (9%) / Undisclosed 1 (3%)
Program: Undergrad (Year 2–5): 20 (63%) / Grad: 12 (37%)
 Undergrad Year: Y2 1 (3%) / Y3 14 (44%) / Y4 4 (13%) / Y5 1 (3%)
Race: White 22 (69%) / Asian 4 (13%) / Black 3 (9%)
 Hispanic 1 (3%) / Mixed 2 (6%)
Disability or Mental Health condition^b: 11 (34%)
Age Range: Undergrad 20–33 / Grad 23–41

^aIncludes one transgender woman.

^bADHD 6; Autism 2; Mental Health 6. 1 unspecified disability. 4 had multiple or overlapping.

Table 1: Participant Demographics (N=32 respondents)

included one all disabled men group of 4, two groups with two of four members disabled, and a group of 4 with one disabled. No gender or disability status was disclosed to students by the instructor, only group numbers and meeting times were shared. Reflection group sizes are reported for the actual size of the groups, which do not add up to 34 research participants, as three students opted-out of the research (from the mixed groups).

5 RESULTS

Our overall goal is a qualitative initial evaluation, while also building understanding and theory for why and how in computing settings this kind of collaborative and individual "weekly continuous learning and personal growth sprint" operates and its effects. This can help educators and researchers further experiment and refine the design for this assignment and related practices [40]. We are not asserting all students experienced all of these things, but instead broadly what kinds of experiences and categories of learning happened, and the kinds of causes students reported as facilitating that learning. Each **theme** is **bolded** and needed to have at least three unique students (and thus three supporting quotes) to be included in our results, although for space reasons we included only a few examples balancing space and richness of the student reflections. The student names are pseudonyms with gender e.g., Pat for undisclosed gender. Student quotes are formatted as **colored text** in the results section to show more quotes and enhance readability.

5.1 RQ2: Categories of learning and effects from the reflection groups reported by students

These categories ranged from learning skills, often professional skills (such as communication), to skills and dispositions for improving their ability to improve (such as being more honest with themselves and new ways of thinking and reflecting). Some students even transferred their learning and reported positive impacts in their relationships outside the course, even in their families. .

New habits of reflection, experimentation, and smaller more reachable goals: Students reported reflecting more and seeing value in reflection, which is critical for lifelong learning. For example, Tom: "I learned that it can be helpful to take a moment to reflect on what is going on in your life...I started to think about stuff I could make small improvements on, even If it doesn't feel like much. Small steps help.", and Nick: "The ability to reflect on past habits, and just thinking about how I could improve those habits helped me quite a bit. I just do this more than I used to."

Students reported more experimenting and persisting through initial failure; for example, Jeff: "By learning from my mistakes

throughout the semester, I was able to get more work done and plan my days out more efficiently." and John: "It is not perfect yet, but making sure I get good sleep has helped me a lot to stay productive. Figured out what works for me/what doesn't." Students such as Pat also reported setting smaller goals where "I feel positively, b/c it let me make progress even if to a smaller degree..." and Jeff: "I worked on my patience, motivation, and setting smaller, more reachable goals so they feel more rewarding."

More ready and open for group work: Students reported feeling more comfortable in groups, improved ability to share honestly in groups, and increased safety. For example, Ed: "I learned how to be more honest with others around me in a way beneficial for me. I've begun feeling more comfortable in group settings.", James: "it made me more open to working with others in the future.", and Scott on: "honestly shar[ing] goals with my peers...I am normally an individually motivated person, so it was challenging at first."

Future professional collaboration skills: Students reported improved future professional and workplace skills. For example, Barry: "Being able to grow as a team helped me know what makes a team work, and be able to apply this to my life to work with people.", and Dan: "I have learned to become a better team member by knowing how to discuss my goals and check in on the goals of others, which benefits me and any teams I am on."

Generative AI skills and what GenAI means for career: Students reported improvement in their understanding of GenAI and what it means for their careers. For example, Brian saw practical improvements: "Improved at using generative AI to learn...Been more effectively able to use AI for a variety of different things...Using AI more often for assignments/summaries." For what it means for their careers, Ed wrote: "The others in my team were very helpful in my understanding of how others view the 'AI hijacking' of tech...Each of my team members encountered this change in a unique way that possibly challenged their career goals."

More self-awareness and self-honesty Students reported learning to be more self-aware and honest; for example, Blake: "More openness with myself and a better state of mind due to self improvement...I improved with my openness and self inspection.", Harry: "I improved at being honest with myself", and Cal: "...more self aware of my weekly habits and how they could improve"

Feeling not alone, having shared experiences or struggles: Students reported feeling less alone from a sense of shared experiences. For example, Barry: "I also felt more positive about the week, whether or not I was productive. If I was productive it felt good to share, if I wasn't it was good to know that I wasn't alone with falling short", and James: "Discussing our concerns for working in this field created a space where we felt understood or shared a similar experience, helping stifle anxiety and stress".

Improved communication skills: Students reported improved communication skills in general, for example, James: "I think it allowed me to have a clearer mind in discussions, so that I could accurately express and explain my thoughts without issue." and Nick: "I just became a better communicator by the end of the semester." Students also reported **improving at communicating struggles**, for example, Tom: "It helped us ask out other group mates for help on stuff you struggle with." Students also reported **improved skills for listening and empathy**, for example, Andy:

"My ability to listen and comprehend what other people are saying improved over this semester", and Rachel: "I think it makes me better at listening and understanding. It's a lot easier to help others if you can improve that skill, I think." Students also reported **recognizing general deeper truths about communication and social connection**, for example, Rob: "Provided a good example of how conversation and a space to share and reflect can cause adjustments and improvements to individual and group systems", and Jordan: "Trust is a choice you make."

Improved relationships outside the reflection group: Students' improved communication and other skills led to improved relationships, transferring beyond the course; for example, Andy: "This had a great impact on the people that I'm around. My family said that they noticed that I was talking to them more and also I got compliments on my public speaking skills.", Jason: "It affected my personal relationships overall because it allowed me to learn to be more open.", Ed: "It's improved my relationships in my fraternity, family, and roommates.", Peter: "My family felt more valued and respected when they could tell that I was putting more effort into being aware and 'present' in the moment."

Planning and time management: Students reported improved time management and planning skills. For example, Tom: "Improved time management...It helped me make progress on something I have struggled with all my life.", Will: "Improved organizational and improved meeting deadlines consistently.", and Ken: "This improved my stress level as my time management got better and help me have more time to spend with other people."

5.2 RQ3: According to students, what helped students a) improve, and b) improve their ability to improve as a reflection group?

The themes here included a sense of community and shared experiences, and growing trust and safety to be vulnerable. This led to more openness and honesty, which enabled feedback loops including sharing goals, problems, solutions, and advice. Students also mentioned basic features of the reflection groups assignment, such as the weekly group meeting agenda and submitting post-meeting individual reflection question responses, as well as other causes and helpful factors.

Trust and safety: Students reported feelings of trust and psychological safety, as a result of the teams, helped them improve their ability to improve as a team and individually. Students reported feeling more comfortable sharing, such as Peter: "This improvement happened because of the development of trust between team members and the realization by everyone that this was a safe place to share thoughts, even if they weren't fully developed." Lauren: "It gave a safe space for discussion. We could learn from one another and help each other." Peter: "The development of personal vulnerability helped to create a shared CoP for the class, for school, and really (more broadly) life. I would not have been open to this idea absent the class." Students reported that, as a result of the safe space in the groups, they were able to have more open dialogue, such as Frank: "The team created a safe space to share honest feedback. Their encouragement and willingness to listen helped me feel comfortable contributing and improving.", and Andy: "We were able to talk about many issues and often have different opinions. This did

not stop my learning process and encourage me to talk more...This happened because school created a safe space for us to talk."

Openness, honesty, and genuineness: Students reported that trust and safety led to more openness, honesty, and genuineness, which helped them improve. James said they "Broke down the barriers to allow more direct and open conversation, and developed trust to express our own thoughts without judgement." Cal: "Through genuine conversations and interest with each other. We all felt like we were on the same page and didn't have the need to impress anyone." Nick: "We were better communicators by the end of the semester, we got better at improving as a result. We could just be more open with one another."

Sharing Problems: Students also noted improvement through help from the team sharing their problems. For example, Jeff: "As the semester went on, we became more comfortable with each other, allowing us to share more about our successes and failures. We were then able to examine these things and try to find a solution to solve these problems.", and Brian: "Other people had similar problems and it's easier to get through a problem together."

Sharing Goals: Sharing goals was another theme of improvement that students described. For example, Jordan: "I think as we felt more comfortable with each other, we... became downright shameless in 'stealing' each other's goal to try to improve on them ourselves.", and Jess: "Over the course of the semester we each shared personal goals and roadblocks."

Sharing Progress: Students proposed that sharing their progress with their team helped them individually and together. For example, John: "Discussion and reflection about other team member's habits/things that worked for them has helped me", and Barry: "I also felt more positive about the week, whether or not I was productive. If I was productive it felt good to share, if I wasn't it was good to know that I wasn't alone with falling short."

Receiving useful help through feedback and encouragement: Students also highlighted that they improved by receiving useful help through feedback and encouragement from their classmates. Examples included Rob, "There was a plethora of positive feedback, excellent conversations and brainstorming for growth" and Peter, "It was productive to be able to get feedback from people who didn't know me well, but where I was still comfortable enough to be relatively vulnerable."

Weekly individual reflection assignment: After each weekly reflection group meeting, students were instructed to submit individual written responses to the reflection questions. Students reported this helped them improve. For example, Nick mentioned the individual reflection assignments: "Just reflecting more on my habits helped my ability to improve, improve. This was caused due to the reflection assignments, and our weekly meetings." and Frank "I believe I got better at reflecting on what worked and what did not after each meeting... It happened because I was given consistent feedback from the professor". The only feedback given on this from the professor was the grade for just submitting the reflection.

Weekly group meeting agenda: The agenda included writing bullet points on recurring questions about student improvement and goal setting, then discussing. Students reported specifically these helped, for example Cal: "It was helpful for everyone in our group to brainstorm some bullet points of on what they could've done better, especially over the course of the semester to see repeating

habits.", Dan: "Being required to have the meetings and document our responses made this happen. Having our goals in writing made me more likely to stick to them...Having team members read my goals made me even more incentivized to write them down and stick to them.", and Rich: Just reflecting out loud to other people, and gaining feedback along with that helped a lot, especially when a lot of my improvements I was working on, a lot of my team members were also working or worked on."

6 LIMITATIONS

We limit our claims and results to the course and student population in our dataset. Given the novelty of our reflection design, it can likely be improved and potentially personalized for different student populations or course contexts. Different or more scaffolding might be required particularly for more novice computing students. Recall and/or social desirability bias may have influenced student responses; our assignment was graded for completion only, which reduces that bias [37]. Due to space we reported quotes from 26 unique students, with with Student quotes

7 DISCUSSION

We contribute reflection groups, a mostly outside-of-class scaffolded assignment for lifelong learning, communication, and other professional skills. Our analysis revealed, even in our online setting, learners improved their reflection and communication skills, including listening and being vulnerable by sharing their struggles and problems. We found broad categories of learning and specific ways students improved their ability to improve as a team. Students created better improvement habits that helped them reach their goals, collaborate better, and develop professional skills. Students also developed emotional skills such as better relationships, self-awareness, and learned to prioritize their wellness. Students reported reflection groups helped them learn because they scaffolded an environment where students felt safe and could trust one another. This led to openness and vulnerability, enabling feedback that helped each other reach goals. This was scaffolded via the in-class reflection group icebreaker activity. This constitutes initial evidence the group design supports breadth and depth in learning professional skills, even in an online setting.

A strength of our work is that the reflection questions' flexibility supported learning for a broad set of professional skills and other outcomes. Many of the learning categories from the reflection groups directly covered professional skills and dispositions in the ACM CS2023 Computing Curriculum, such as Adaptable, Collaborative, and Self-directed [36]. While skills are important, dispositions are the willingness and intent to apply skills; they enable consistent behavior [36]. Student reflections aligned with dispositional changes in becoming more self-aware and honest, reflective, and more comfortable with being vulnerable in a group setting, particularly given explicit reports of transfer. Future work should further investigate changes in dispositions and professional skills with quantitative measures. Future work will also analyze the 400+ weekly reflections, which may reveal how learning developed over time. Our work also informs what skills and dispositions to target given limited student time for surveys and evaluation.

Aggregate education and industry studies consistently identify professional skills as important for computing careers, yet also find that graduates are under-prepared in areas like communication and lifelong learning [28, 29, 45]. This shows group projects and other existing professional skills learning in computing are not enough. As our initial promising results aligned with that need, future work should eagerly explore and evaluate reflection groups in different courses and with different student populations. It is possible that in-person groups, with the same intentional scaffolding, could achieve comparable or even better results.

Reflection groups did also support student connection, especially important as LLM help-seeking erodes student and course communities [31]. Community-enhancing interventions like reflection groups are sorely needed to counter that trend. We understand readers may be skeptical that students would meet outside of class. Regardless, our empirical results suggest that carefully fostering student connection, and designing group meetings to center students' agency in choosing what they want to improve, can create communities where students choose to participate because they feel less alone, and receive useful help through peer feedback and encouragement. To adopt reflection groups in a course, an instructor uses 50 to 60 minutes of class time on the icebreaker activity, and adds a weekly assignment for outside-of-class group meetings and a graded-for-completion individual reflection submission ([anonymized materials](#)). Future work should evaluate instructor burden and ideas like integrating added "course-specific" questions.

The success of reflection groups in our online setting among professional learners raises the possibility of integrating professionals into undergraduate computing education, where both groups benefit. In our course, online working professional MS students also reported consistently positive effects, including in their family lives. Could networks of students and professionals reflecting together contribute to computing professional practice becoming more reflective, vulnerable, and open? Past work and even industry research such as Google's re:Work study [1] has found that psychological safety is the top factor for software team effectiveness. Cultural change in computing and STEM is a wicked problem [15] - one way to approach it is to make change at a small scale, like in a small reflection group, then grow that change into larger groups.

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